

Deliverable 2: Project Proposal

1. **Introduction:** Formula 1 is a globally recognized motorsport defined by engineering and high-speed strategy. The immense amount of data generated from the first lap to the checkered flag presents a rich opportunity for data analysts. The project will use two datasets of the 2024 and 2025 Formula 1 seasons.
2. **Purpose of the project:** The main purpose of the Project is to apply data techniques that will help us analyze the information of the 2024 and 2025 Formula one season included in both datasets and find out if FIA wind tunnel regulations helped smaller teams to increase their performance and equal competition between Teams
3. **Significance:** The project is significant because it provides an approach to different teams results between two different seasons and how relevant wind tunnel hours are for development of F1 cars
4. **Primary Objective:** The primary Objective of the Project is to demonstrate that the new Formula 1 regulations over the hours of wind tunnel every team had over a season, can affect their performance in future seasons.
5. **Secondary Objectives:**
 - Compare drivers results and points over two different Formula one seasons
 - Compare points achieved by every Team during the 2024 and 2025 Formula one seasons
 - Identify the number of wind tunnel Hours each Team have available during the 2025 Formula one season.
 - Compare The income of every Formula one team at the end of the 2024 season
6. **Approach:** This project will follow a structured data analysis, beginning with data gathering and cleaning. The data, which is provided in CSV format, will be imported to a new project in RStudio in the programming language R. This will permit me to analyze and model data into graphics that represents the different data values from the csv file

7. Tools and resources:

- Dataset: F1_race_traces and F1_wind_tunnel
- Programming Language: R
- Libraries: Tidyverse, dplyr and tidyr
- Environment: RStudio (Rmarkdown)
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8. Timeline:

- October 19th: Data acquisition and cleaning
- October 26th: Exploratory data analysis (EDA) and initial hypothesis testing.
- November 9th- Final draft report
- November 16th - Final Deliverable
- November 17th-21st- Final presentation
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9. Deliverables:

- A comprehensive Rmarkdown report with code visualizations and interpretations
- An HTML file with all the outputs summarizing the key findings

10. **Success Criteria:** Success will be measured in how well-structured and organized the datasets are organized and if the representation for the project clearly represents the data from the original dataset.

11. Potential Challenges:

- Data Quality: The dataset may contain some "ugly data" such as inconsistent formatting, or missing values that require significant cleaning
- How specific is the data: The dataset might not include all desired variables, such as detailed telemetry or specific weather conditions for each race, which could limit the depth of the analysis.

12. Limitations:

- The analysis is limited to the data provided in the F1 Race Traces - 2025 dataset and may not include all factors that influence race results.
- The information provided may not be enough because the 2025 F1 season is not over yet

13. **Conclusion:** This project proposes an exploration of the 2025 Formula 1 season. By analyzing the "F1 Race Traces - 2025" dataset, we aim to provide valuable insights into the performance of F1 drivers and teams.

14. References:

- Trotman, J. (2025). F1 Race Traces — 2025. Kaggle.
<https://www.kaggle.com/code/jtrotman/f1-race-traces-2025>